



NestNetR: Deep-learning based analysis of breeding behaviour from light and temperature recordings in R

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Abstract

1. Advances in biologging have generated large and increasingly complex datasets, creating unprecedented opportunities to study animal behaviour across individuals, populations, and species, while simultaneously increasing the need for objective and scalable analytical approaches.
2. NestNetR is an open-source R package that automates the analysis of avian breeding behaviour from light-level geolocator data using deep learning. The package integrates data preprocessing, model definition, and behavioural classification through convolutional neural networks (CNNs), providing a reproducible and modular workflow.
3. To demonstrate its application, we trained a CNN on geolocator data from Ruddy Turnstones (*Arenaria interpres*), an Arctic-breeding shorebird. The model achieved high predictive accuracy (micro F1-score = 0.92 ± 0.01), effectively distinguishing between daily incubation, brooding, and other unclassified activity from combined light and temperature time series.
4. NestNetR offers an accessible framework for inferring breeding behaviour from biollogger data, reducing manual effort and observer bias. It extends the use of light-level geolocator data beyond migration research and facilitates large-scale, objective analyses of avian breeding behaviour and behavioural classification through convolutional neural networks (CNNs) providing a reproducible and modular workflow.

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Introduction

Biologging technologies are increasingly used to study animal behaviour by recording movements and local environmental conditions in unprecedented detail, enabling objective, large-scale analyses across a wide range of taxa and environments (Kays et al., 2015; Rutz & Hays, 2009). In parallel, the availability of biologging data has expanded rapidly through public repositories and long-term monitoring programmes (Nathan et al., 2022; Wüest et al., 2020). Together, these developments provide new opportunities to investigate behavioural processes at scales that were previously inaccessible (Rutz & Hays, 2009; Wilmers et al., 2015).

Light-level geolocators, in particular, represent a widely used class of biologging devices in avian research, providing information not only on movement but also on behavioural states through recorded light and temperature data (Burger et al., 2012; Gosbell et al., 2012; Lisovski et al., 2016; Loktionov et al., 2015; Schaub et al., 2019). Except for passerines, these loggers are typically attached to the bird's leg, recording light exposure and temperature close to the surface. In this context, polar environments, characterised by continuous daylight and the absence of twilight, represent a particularly informative application system, as small fluctuations in light intensity and temperature can be used to infer breeding-related behaviours (Bulla et al., 2016; Burger et al., 2012; Gosbell et al., 2012; Lisovski et al., 2016). However, the manual identification of behavioural states is labour-intensive and often highly subjective, limiting comparability across studies (Williams et al., 2020). To overcome these limitations that are applicable to many behavioural datasets, automated and standardised classification methods using machine learning approaches are increasingly being explored, offering objective and scalable solutions (Keitt & Abelson, 2021; Nathan et al., 2022).

Building on these advances, deep learning represents a specialised and powerful method for objective analysis of complex ecological time-series (Borowiec et al., 2022; Brodrick et al., 2019). Here, we present a tool and workflow that applies a deep neural network to classify daily breeding behaviour into three states – incubation, brooding and unclassified activity – from geocator-derived light and/or temperature time series. We demonstrate the workflow using a dataset of Ruddy Turnstones (*Arenaria*

interpres), a long-distance migratory shorebird breeding in the high Arctic. This system serves as an illustrative example of how deep learning can be used to analyse breeding behaviour from light and temperature recordings, particularly under polar summer conditions where continuous daylight and limited field access make conventional behavioural monitoring challenging. Overall, the workflow aims to establish a consistent, data-driven framework for identifying breeding behaviour from bilogger time series, enabling reproducible inference across individuals, populations, and years.

R-package NestNetR

Overview

NestNetR is an R package developed to identify and classify avian breeding behaviour from time series of light intensity and, if available, temperature measurements using a deep learning framework. The package provides a complete and reproducible, 3-step workflow that integrates data preprocessing, model training, and automated behavioural classification using the trained model (Tab. 1). Breeding states are inferred using a convolutional neural network (CNN), which is well suited for detecting characteristic temporal structures across multiple scales via so-called stacked one-dimensional convolutional and pooling operations (O'shea & Nash, 2015). Models are implemented in R via Keras with a TensorFlow backend, which together provide open-source libraries for defining, training and deploying deep neural networks within the R environment (Chollet, 2015; Kalinowski et al., 2025).

The resulting model distinguishes three categories of breeding behaviour: incubation, brooding, and random (unclassified) activities. Incubation is characterised by extended periods of elevated and stable temperature and low or absent light, reflecting continuous nest attendance (Gosbell et al., 2012). Brooding consists of shorter, more frequent temperature drops and light increases caused by parental exchanges and chick care (Loktionov et al., 2015), while random activity represents irregular temperature-light dynamics typical of unclassified breeding or non-breeding states such as mating, foraging or resting.

Table 1 – Description of functions available in the NestNetR package

Function name	Description	Category
<code>read_light</code>	Reads and parses raw light-intensity data from biologgers, returning timestamps and log-transformed, normalised light values.	input
<code>read_deg</code>	Reads temperature data from .deg files and returns daily minimum (Tmin) and maximum temperatures (Tmax).	
<code>set_breeding_period</code>	Determines the breeding-period range (start & end) either automatically from twilight files or manually through an interactive selection tool.	preprocessing
<code>preprocessing</code>	Cleans, rescales and segments bilogger data (light, Tmin, Tmax) into standardised periods for model input.	
<code>create_breeding_data_list</code>	Automates the full data preparation workflow across multiple individuals, combining reading, trimming and segmentation.	
<code>create_trainingdata</code>	Opens an interactive GUI to manually label randomly chosen segments from all available individuals as incubation, brooding or random for model training.	training data
<code>classify_breeding_behaviour</code>	Applies either a pre-trained CNN model or a customly designed model to classify breeding behaviour per segment and outputs predicted probabilities and labels.	application

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89 Step 1: Preprocessing of input data

90 The preprocessing workflow in NestNetR prepares raw light-level geolocator data for
 91 behavioural classification by converting input variables into a uniform, model-ready
 92 time-series format. The functions *read_light* (light intensity) and *read_deg*
 93 (temperature) automatically import raw measurement files in supported formats (e.g.
 94 .lux for light and .deg for temperature data). These data are typically stored as plain
 95 text files by the manufacturer or archival-tag software, and can therefore also be read
 96 directly into R using the base function *read.table*.

97 Because raw data typically cover the entire migration cycle of an individual, the dataset
 98 must first be trimmed to the breeding period by using the *set_breeding_period*
 99 function, which identifies the start and end of the polar breeding phase either
 100 automatically, based on an additional available twilight file (.twl, `auto = TRUE`; s.

GeoLight package by Lisovski & Hahn, 2012, or *TwGeos* package by Wotherspoon et al., 2016), or manually through user selection in an interactive tool (auto = FALSE).

Within the defined breeding period, the *preprocessing* function assembles and segments biollogger data into model-ready inputs; the function first sub-sets all available variable records to the length of the breeding window. Here, light data serve as the reference timeline, and all additional variables – typically recorded at lower temporal resolution – are interpolated to match this scale within their respective sampling interval. The data are then divided into periods according to user-defined length (e.g. one, two, or more days) and potential overlap (e.g. zero days to one day). Each period is stored as a separate list element containing the corresponding variable data for that time span. To ensure comparability between variables, all input channels are scaled to a range of 0 to 1 after clipping values outside the 2.5th–97.5th percentile range of the entire available data, identifying realistic value ranges and minimising the influence of extreme outliers. The resulting input for the neural network therefore consists of an aligned time series of light intensity (light), minimum (Tmin), and maximum temperature (Tmax). However, users are able to include fewer or additional input variables.

Step 2: Model definition (optional)

NestNetR provides the option for users to either apply a pre-trained base model (developed using Ruddy Turnstone data with fixed 24-hour input periods) or define a custom model. If the pre-trained model is used, this step can be omitted. The workflow presented here follows the structure of the base model but allows full flexibility in the selection of period length, input variables, model architecture, and training strategy. Model definition is implemented using functions from the *keras3* package (Kalinowski et al., 2025).

Step 3: Model application

Using the *classify_breeding_behaviour* function in NestNetR, users can either apply the integrated Ruddy Turnstone model or use a custom-trained model from the previous step. The function takes preprocessed data and a trained *keras* neural network model as input to classify breeding behaviour for the specified breeding period. For each time step, the model assigns one of the three behavioural categories

(incubation, brooding, random) based on the class probabilities produced by the network.

The output includes both a complete time series of daily classifications. These results allow users to infer key breeding parameters directly from geolocator data and to compare behavioural timing and breeding success across individuals or populations.

Ruddy Turnstone model

Overview of the Dataset

To demonstrate the applicability of our workflow under realistic ecological conditions, we used data from Ruddy Turnstones (*Arenaria interpres*) – a long-distance migratory shorebird breeding in the high Arctic. The dataset comprises geolocator records since 2009, resulting in 372 individual tracks from birds of different populations (King Island, Tasmania, South Australia and Victoria). Ruddy Turnstones typically exhibit biparental incubation, with both parents alternating bouts of continuous nest attendance (Cramp et al., 1983; Gosbell et al., 2012). These alternating on- and off-nest phases create consistent temporal patterns in light and temperature recordings that reflect breeding behaviours such as incubation and brooding (Gosbell et al., 2012; Lisovski et al., 2016).

Neural Network Architecture

The final convolutional neural network (CNN) architecture consists of an input layer receiving three channels (Light, Tmin, Tmax) with 288 consecutive 5-minute intervals per day, corresponding to a selected, full 24-hour period. Light data were log-transformed prior to rescaling to enhance the representation of low-light variation. To identify the most suitable model, a systematic hyperparameter optimisation procedure was implemented during model development, comparing multiple neural network architectures and structural settings (see Supplementary Material A). This tuning procedure, implemented using the *kerastuneR* package (Abdullayev, 2024), explored different composition of input variables, varying sizes of training data (10 – 100%), alternative distributions of behavioural classes within the training data (equal, ...), and a wide range of architectural hyperparameters. Specifically, a custom model-building function was defined to explore combinations of key architectural components,

including the number of filters (32–480, step size = 16) and kernel sizes (2–20, step size = 2) in the convolutional layers, pooling sizes (4–20, step size = 2), dropout rates (0.1–0.5, step size = 0.1), and dense layer units (128–480, step size = 16). Optimisation was performed using Hyperband tuning (Li et al., 2018), which adaptively allocates computational resources to the most promising configurations. Three Hyperband iterations were conducted, and performance was evaluated using micro F1-score to identify the most robust and generalisable model parameters. The results of this optimisation and comparative tests formed the basis for selecting the final CNN architecture applied in all subsequent analyses.

Two consecutive one-dimensional convolutional layers extract local temporal features from the input data. Convolutional layers identify recurring temporal patterns in light and temperature data, with the first layer applying 400 filters with a kernel size of 12, followed by an average-pooling layer (pool size = 12, padding = "same"). Pooling reduced temporal resolution while retaining behaviour-relevant signal structure, improving computational efficiency (Akhtar & Ragavendran, 2020). The second convolutional layer applies 288 filters with a kernel size of 6, again followed by an average-pooling layer (pool size = 6, padding = "same"). All convolutional layers use ReLU activation to introduce non-linearity. The extracted features are flattened and passed through a dropout layer (rate = 0.2) to prevent overfitting, followed by a fully connected dense layer with 160 neurons and ReLU activation to integrate extracted features across the temporal window. The final output layer comprises three units corresponding to the behavioural classes (brooding, incubation, random) and uses a softmax activation function to produce class probabilities that sum to one.

Model training

For model training, samples were selected from the entire available breeding dataset. Each sample represented a selected, full 24-hour period aligned to calendar days, starting and ending at midnight (00:00–24:00, UTC). Daily periods were recorded at a temporal resolution of five minutes, resulting in 288 consecutive time steps per sample (24 hours × 12 five-minute intervals per hour). Each sample therefore comprised a full daily sequence of light, T_{min}, and T_{max} values, serving as a single input instance for the model. Manually classified samples were partitioned into 80% training and 20% test data (Kuhn & Max, 2008). The resulting training dataset consisted of 1,366 daily

samples (727 random, 481 incubation and 158 brooding samples) and was intentionally kept imbalanced in terms of class representation, reflecting the natural occurring frequency of breeding behaviors.

The model was trained using five-fold cross-validation to evaluate its stability and general performance across subsets of the dataset (functions included in the *caret* package by Kuhn & Max, 2008). Each fold used 80% of the data for training and 20% for validation. Categorical cross-entropy was used as a loss function. The model was optimised using the Adam algorithm with a learning rate of 0.001, $\beta_1 = 0.9$ and $\beta_2 = 0.999$. To account for the natural imbalance among behavioural classes, class weights were applied during training, giving higher importance to rarer (brooding) and analytically more important categories (incubation; brooding = 2.5, incubation = 2, random = 0.8). Training was performed for up to 50 epochs with a batch size of 64, and included early stopping criteria (patience = 8 epochs without improvement) and adaptive learning-rate reduction (factor = 0.5, minimum learning rate = $1e-6$) to prevent overfitting and improve convergence.

Model performance was monitored on the validation data using F1-score, with mean and standard deviation across folds summarising the final evaluation. These metrics were selected based on established recommendations for evaluating multi-class classification models, as they jointly assess both overall and class-specific performance (Grandini et al., 2020; Terven et al., 2025). Model checkpoints were saved for the best performing epoch in each fold.

Model evaluation

For final evaluation, the model achieving the highest validation micro F1-score during cross-validation was selected as the best-performing configuration and subsequently re-trained on the entire training dataset using identical settings to maximise predictive generality. Model performance was then assessed on the held-out test dataset ($n = 331$ daily samples) and summarised using F1-score (micro and macro), categorical accuracy and area under the curve (AUC). The final model achieved an F1-score (micro) of 0.92 ± 0.01 , an F1-score (macro) of 0.84 ± 0.01 , a categorical accuracy of 0.92 ± 0.01 , and an AUC of 0.97 ± 0.00 on the independent test dataset. Exemplary results are shown in Figure 1.

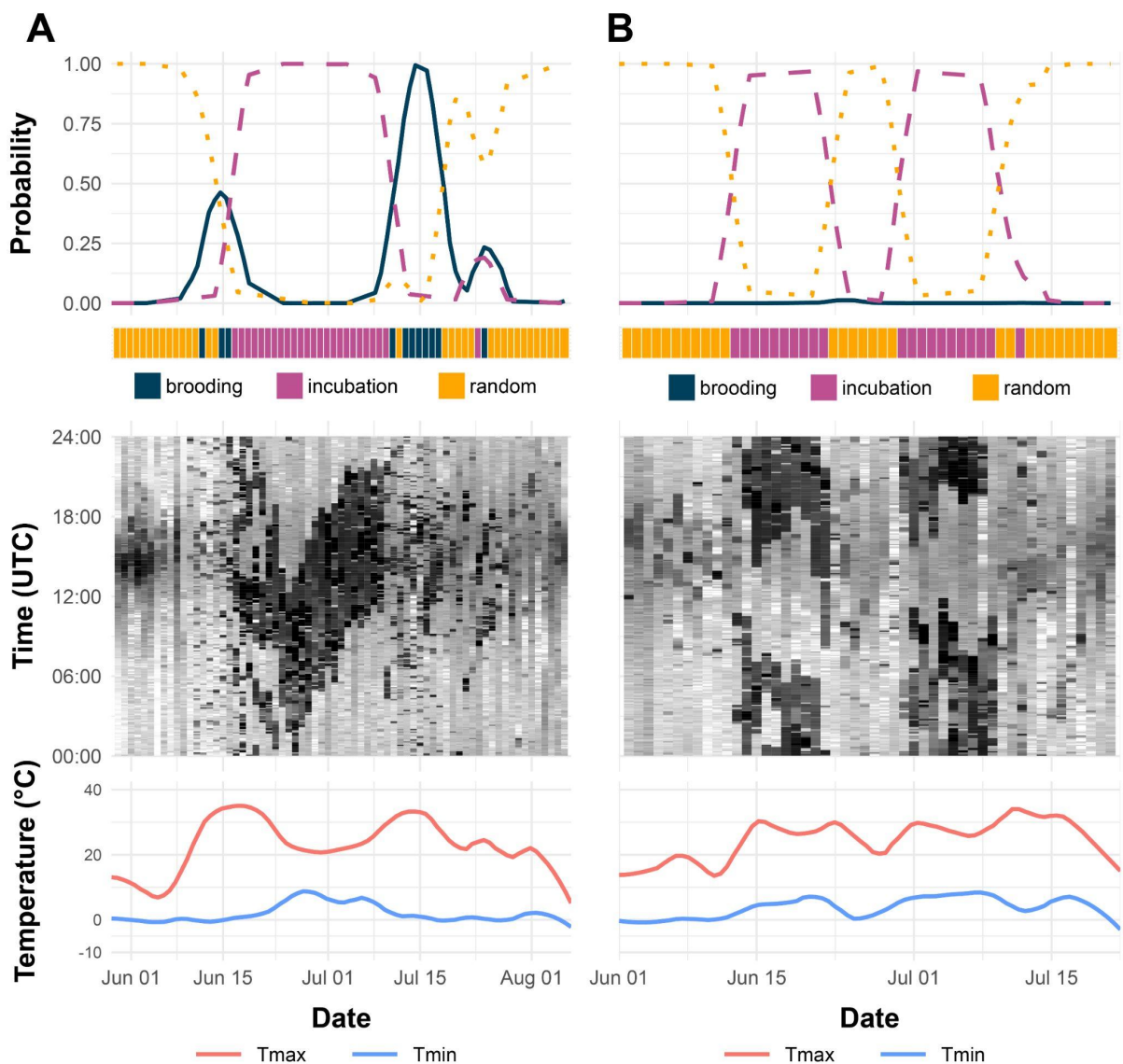


Figure 1 – Example model predictions and bilogger data for two Ruddy Turnstones. For each panel (A & B), the top row shows smoothed class probabilities (span = 0.2) for each behavioural category (brooding, incubation, random activity) over time, with predicted classes (highest probability) indicated below the x-axis. The middle row displays light intensity (lux) derived from geolocator data for each individual, where shading from dark (low) to white (high) represents recorded light intensity over time. The bottom row shows smoothed (span = 0.2) recorded minimum (blue) and maximum temperatures (red) for each logger.

Application to other species

Most shorebird species exhibit biparental care (Bulla et al., 2015). Although the duration of incubation bouts may vary among species, the overall structure and temporal patterns of light–dark cycles and temperature fluctuations during incubation are largely comparable. We therefore applied the Ruddy Turnstone model to available datasets from three additional Arctic-breeding shorebird species – the Great Knot

(*Calidris tenuirostris*; Lisovski et al., 2016), Curlew Sandpiper (*Calidris ferruginea*; Lisovski et al., 2020a), and Red-necked stint (*Calidris ruficollis*; Lisovski et al., 2020b) – to evaluate its potential generality across related taxa.

Model performance was assessed on preclassified evaluation datasets of each species. For the Great Knot dataset (n = 352 daily samples), the Turnstone model achieved an F1-score (micro) of 0.84, an F1-score (macro) of 0.75, a categorical accuracy of 0.84, and an AUC of 0.88. For the Red-Necked Stint dataset (n = 247 daily samples), the Turnstone model achieved an F1-score (micro) of 0.99, an F1-score (macro) 0.97, a categorical accuracy of 0.97, and an AUC of 0.99. For the Curlew Sandpiper dataset (n = 534 daily samples), the Turnstone model achieved an F1-score (micro) of 0.97, an F1-score (macro) 0.97, a categorical accuracy of 0.99, and an AUC of 0.99.

Discussion

Here, we present a deep learning approach to analyse breeding behaviour from light-level geolocator data. By combining derived light and temperature data exemplarily from a long-term dataset of Ruddy Turnstones with a convolutional neural network, we demonstrate that complex behavioural patterns can be objectively identified under polar conditions, where direct field observations are rarely feasible. The developed Ruddy Turnstone model showed high predictive performance, accurately distinguishing incubation from brooding and unclassified breeding activities .

With NestNetR, we provide an R package offering a reproducible and transferable framework for behavioural inference that is specifically suitable for polar environments, where the absence of pronounced day-night cycles allows subtle light and temperature fluctuations to be exploited for behavioural classification. At the same time, the underlying approach is not inherently restricted to polar systems, and there is clear potential for its application in other regions and species, including those experiencing regular diurnal light-dark cycles. In such systems, day-night variation may introduce additional signal structure or noise, but may also provide complementary information that could be leveraged through adapted model architectures or training strategies. The approach reduces manual effort and observer bias while maintaining high accuracy, and illustrates how the use of geolocator-derived data can extend their

application beyond migration studies. With further methodological refinement and region-specific tuning, including post-processing or alternative temporal aggregation strategies, classification performance may be improved even further, while application could also be extended to non-polar environments.

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